

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical
Practical - VII

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Hash Table:

Write a program to:

- a. Implement a hash table with separate chaining for collision handling.

Code:

```
size = 5
table = [ [] for _ in range(size) ]

def insert( key,value ):
    table[ (key%size) ].append(( key,value ))

insert(10,"Amit")
insert(12,"Raj")
insert(15,"Akuv")
insert(20,"Jay")

print(table)
```

Output:

```
= RESTART: C:/Users/IT-47/AppData/Local/Programs/Python/.
[[ (10, 'Amit'), (15, 'Akuv'), (20, 'Jay') ], [], [(12, 'Raj')], [], []]
```

- b. Store and retrieve data from the hash table.

Code:

```
size = 5
table = [ [] for _ in range(size) ]
```

```
def insert( key,value ):
    table[ ( key%size) ].append(( key,value ))
```

```
def search( key ):
    for k, v in table [ key % size ]:
        if k == key:
            return v
    return "Not Found"
```

```
insert(10,"Amit")
insert(12,"Raj")
insert(15,"Akuv")
insert(20,"Jay")
```

```
print(table)
```

```
key = int(input("Enter Key"))
print(search(key))
```

```
print("Amit 44")
```

Output:

```
[[ (10, 'Amit'), (15, 'Akuv'), (20, 'Jay')], [], [(12, 'Raj')], [], []]
```

```
Enter Key10
```

```
Amit
```

```
Amit 44
```

Curiosity Questions:

1. In a BST, which side (left or right) has **smaller numbers** than the root?
2. If we insert numbers 10, 20, 5 into a BST, which number will be at the root?
3. What will the inorder traversal of a BST give — sorted numbers or random numbers?
4. Where will the **largest number** in a BST always be found?
5. If the root of a BST is 50 and we search for 70, will we go left or right? Why?
6. If a BST has only **one node**, what is its inorder traversal?
7. Can we create a BST from characters like A, B, C instead of numbers?
8. If we insert numbers 5, 10, 15, 20 in order, will the tree be balanced or like a straight line?
9. In a BST, do we check the left child first or the right child first in inorder traversal?
10. If we delete the root of a BST, will the tree disappear?